INCH-POUND
MIL-C-19978/20A
18 June 2001
SUPERSEDING
MIL-C-19978/20
30 August 1974

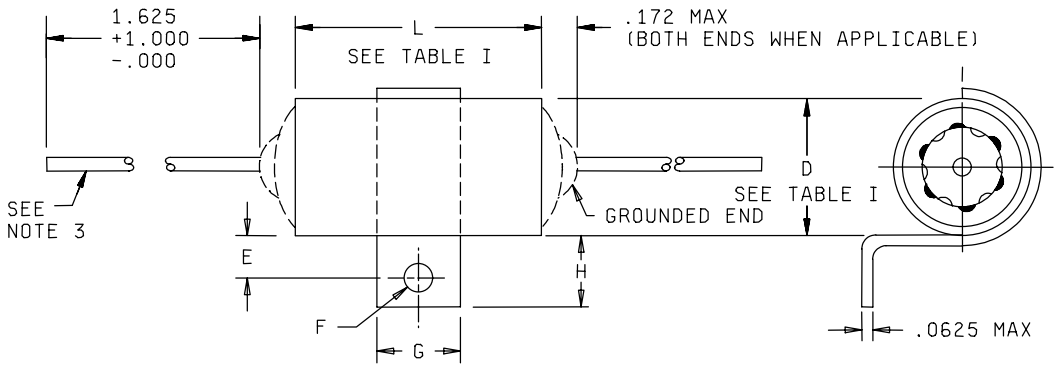
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TANGENTIAL RETAINER, TUBULAR (UNINSULATED), (HERMETICALLY SEALED IN METAL CASES), STYLE CQ10

INACTIVE FOR NEW DESIGN
AFTER 30 AUGUST 1974

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-19978.



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.032	0.81	.172	4.37	.400	10.16
.004	0.10	.062	1.57	.188	4.78	.438	11.13
.005	0.13	.0625	1.59	.235	5.97	.500	12.70
.025	0.64	.144	3.66	.250	6.35	1.000	25.40
.031	0.79	.156	3.96	.312	7.92	1.625	41.28

Case dimension D	H±.062	G±.062	E±.031	F±.005
Less than .500	.312	.250	.188	.144
.500 and up	.438	.500	.250	.156

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Leads shall be of tinned solid wire, .025 (No. 22 AWG) for cases .235 and .312 in diameter; and .032 (No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameters shall be + .004, -.001.

FIGURE 1. Style CQ10 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

TABLE I. Style CQ10 capacitors.

Part or Identifying Number (PIN) 1/	DC voltage rating (volts)	Capacitance (µF)	Case dimensions (inches) 2/		
			D+.010 -.020	Circuit 1 L ±.015	Circuit 3 L ±.015
CQ10A-KB333K3	100	.033	0.235	0.750	0.688
CQ10A-KB823K3	100	.082	0.312	0.875	0.812
CQ10A-KB104K3	100	.10	0.312	0.875	0.812
CQ10A-KB124K3	100	.12	0.400	0.875	0.812
CQ10A-KB154K3	100	.15	0.400	0.875	0.812
CQ10A-KB184K3	100	.18	0.400	1.125	1.062
CQ10A-KB224K3	100	.22	0.400	1.125	1.062
CQ10A-KB274K3	100	.27	0.400	1.375	1.312
CQ10A-KB394K3	100	.39	0.562	1.125	1.062
CQ10A-KB474K3	100	.47	0.562	1.125	1.062
CQ10A-KB684K3	100	.68	0.562	1.375	1.312
CQ10A-KB105K3	100	1.0	0.562	1.875	1.812
CQ10A-KC123K3	200	.012	0.235	0.750	0.688
CQ10A-KC153K3	200	.015	0.235	0.750	0.688
CQ10A-KC223K3	200	.022	0.235	0.750	0.688
CQ10A-KC473K3	200	.047	0.312	0.875	0.812
CQ10A-KC563K3	200	.056	0.312	0.875	0.812
CQ10A-KC683K3	200	.068	0.312	0.875	0.812
CQ10A-KC823K3	200	.082	0.400	0.875	0.812
CQ10A-KC104K3	200	.10	0.400	0.875	0.812
CQ10A-KC124K3	200	.12	0.400	1.125	1.062
CQ10A-KC154K3	200	.15	0.400	1.125	1.062
CQ10A-KC184K3	200	.18	0.400	1.375	1.312
CQ10A-KC224K3	200	.22	0.400	1.375	1.312
CQ10A-KC274K3	200	.27	0.562	1.125	1.062
CQ10A-KC334K3	200	.33	0.562	1.125	1.062
CQ10A-KC474K3	200	.47	0.562	1.375	1.312
CQ10A-KC564K3	200	.56	0.562	1.875	1.812
CQ10A-KC684K3	200	.68	0.562	1.875	1.812
CQ10A-KC824K3	200	.82	0.670	1.875	1.812
CQ10A-KC105K3	200	1.0	0.670	1.875	1.812
CQ10A-KE822K3	400	.0082	0.235	0.750	0.688
CQ10A-KE103K3	400	.010	0.235	0.750	0.688
CQ10A-KE333K3	400	.033	0.312	0.875	0.812
CQ10A-KE393K3	400	.039	0.400	0.875	0.812
CQ10A-KE473K3	400	.047	0.400	0.875	0.812
CQ10A-KE563K3	400	.056	0.400	1.125	1.062
CQ10A-KE683K3	400	.068	0.400	1.125	1.062
CQ10A-KE104K3	400	.10	0.400	1.375	1.312
CQ10A-KE154K3	400	.15	0.562	1.125	1.062
CQ10A-KE184K3	400	.18	0.562	1.375	1.312
CQ10A-KE224K3	400	.22	0.562	1.375	1.312
CQ10A-KE274K3	400	.27	0.562	1.875	1.812
CQ10A-KE334K3	400	.33	0.562	1.875	1.812
CQ10A-KE394K3	400	.39	0.670	1.875	1.812

See footnotes at end of table.

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TABLE I. Style CQ10 capacitors - Continued.

Part or Identifying Number (PIN) 1/	DC voltage rating (volts)	Capacitance (µF)	Case dimensions (inches) 2/		
			D+.010 -.020	Circuit 1 L ±.015	Circuit 3 L ±.015
CQ10A-KE474K3	400	.47	0.670	1.875	1.812
CQ10A-KE564K3	400	.56	0.750	2.125	2.062
CQ10A-KE684K3	400	.68	0.750	2.125	2.062
CQ10A-KE824K3	400	.82	1.000	1.875	1.812
CQ10A-KE105K3	400	1.00	1.000	1.875	1.812
CQ10A-KF102K3	600	.001	0.235	0.750	0.688
CQ10A-KF152K3	600	.0015	0.235	0.750	0.688
CQ10A-KF222K3	600	.0022	0.235	0.750	0.688
CQ10A-KF332K3	600	.0033	0.235	0.750	0.688
CQ10A-KF472K3	600	.0047	0.235	0.750	0.688
CQ10A-KF562K3	600	.0056	0.235	0.750	0.688
CQ10A-KF682K3	600	.0068	0.235	0.750	0.688
CQ10A-KF103K3	600	.010	0.312	0.875	0.812
CQ10A-KF153K3	600	.015	0.312	0.875	0.812
CQ10A-KF223K3	600	.022	0.312	0.875	0.812
CQ10A-KF333K3	600	.033	0.400	0.875	0.812
CQ10A-KF473K3	600	.047	0.400	1.125	1.062
CQ10A-KF683K3	600	.068	0.400	1.375	1.312
CQ10A-KF104K3	600	.10	0.562	1.125	1.062
CQ10A-KF124K3	600	.12	0.562	1.375	1.312
CQ10A-KF154K3	600	.15	0.562	1.875	1.312
CQ10A-KF184K3	600	.18	0.562	1.875	1.812
CQ10A-KF224K3	600	.22	0.562	1.875	1.812
CQ10A-KF274K3	600	.27	0.670	1.875	1.812
CQ10A-KF334K3	600	.33	0.670	1.875	1.812
CQ10A-KF394K3	600	.39	0.750	2.125	2.062
CQ10A-KF474K3	600	.47	0.750	2.125	2.062
CQ10A-KF684K3	600	.68	1.000	1.875	1.812
CQ10A-KG153K3	1000	.015	0.400	0.875	0.812
CQ10A-KG223K3	1000	.022	0.400	1.125	1.062
CQ10A-KG273K3	1000	.027	0.400	1.375	1.312
CQ10A-KG333K3	1000	.033	0.400	1.375	1.312
CQ10A-KG473K3	1000	.047	0.562	1.125	1.062
CQ10A-KG563K3	1000	.056	0.562	1.375	1.312
CQ10A-KG823K3	1000	.082	0.562	1.625	1.562
CQ10A-KG104K3	1000	.10	0.562	1.625	1.562
CQ10A-KG154K3	1000	.15	0.670	1.625	1.562
CQ10A-KG224K3	1000	.22	0.750	2.125	2.062
CQ10A-KG274K3	1000	.27	0.750	2.625	2.562
CQ10A-KG334K3	1000	.33	0.750	2.625	2.562
CQ10A-KG474K3	1000	.47	1.000	2.125	2.062

1/ Complete part number shall include an additional symbol to indicate circuit.
2/ See table II for metric equivalents.

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TABLE II. Millimeter equivalent of decimal inches.

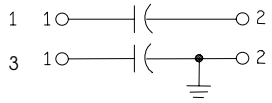
Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.010	0.25	0.670	17.02	1.125	28.58	2.062	52.37
0.015	0.38	0.688	17.48	1.312	33.32	2.125	53.98
0.020	0.51	0.750	19.05	1.375	34.93	2.562	65.07
0.235	5.97	0.812	20.62	1.562	39.67	2.625	66.68
0.312	7.92	0.875	22.23	1.625	41.28		
0.400	10.16	1.000	25.40	1.812	46.02		
0.562	14.27	1.062	26.97	1.875	47.63		

Case material: Nonmagnetic (end seal may be of magnetic material).

Dielectric material: Paper and polyethelene terephthalate (K).

Terminals: Axial-wire lead (see figure 1).

Circuit diagram:



Rated voltage: See table I.

Rated temperature: -65° to +125°C.

Capacitance (nominal): See table I.

Capacitance tolerance: ±10 percent (symbol K).

Vibration grade: 3.

Dissipation factor (DF) (max): 1.0 percent.

Seal: Method 112 of MIL-STD-202, test condition letter A.

Barometric pressure, qualification only: In accordance with MIL-PRF-19978.

Test points:

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

Insulation resistance (IR):

Terminal to terminal: See table III.

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TABLE III. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case: Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-PRF-19978.

Solderability: In accordance with MIL-PRF-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter D.

Stability at low and high temperature:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Capacitance change (max): -10 percent.

High temperature:

Test temperature: +125°C +5°C, -0°C.

Capacitance change (max): +10 percent.

Life:

Capacitance change (max): ±5 percent.

Resistance to soldering heat: In accordance with MIL-PRF-19978.

NOTE: Style CQ10 capacitors are substitutable for style CP10 capacitors formerly covered in MIL-C-25/1K as shown in the following:

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From	To	From	To
MIL-C-25/1K (CP10)	MIL-C-19978/20 (CQ10)	MIL-C-25/1K (CP10)	MIL-C-19978/20 (CQ10)
CP10A-KB333K-	CQ10A-KB333K3-	CP10A-KE334K-	CQ10A-KE334K3-
CP10A-KB823K-	CQ10A-KB823K3-	CP10A-KE394K-	CQ10A-KE394K3-
CP10A-KB104K-	CQ10A-KB104K3-	CP10A-KE474K-	CQ10A-KE474K3-
CP10A-KB124K-	CQ10A-KB124K3-	CP10A-KE564K-	CQ10A-KE564K3-
CP10A-KB154K-	CQ10A-KB154K3-	CP10A-KE684K-	CQ10A-KE684K3-
CP10A-KB184K-	CQ10A-KB184K3-	CP10A-KE824K-	CQ10A-KE824K3-
CP10A-KB224K-	CQ10A-KB224K3-	CP10A-KE105K-	CQ10A-KE105K3-
CP10A-KB274K-	CQ10A-KB274K3-	CP10A-KF102K-	CQ10A-KF102K3-
CP10A-KB394K-	CQ10A-KB394K3-	CP10A-KF152K-	CQ10A-KF152K3-
CP10A-KB474K-	CQ10A-KB474K3-	CP10A-KF222K-	CQ10A-KF222K3-
CP10A-KB684K-	CQ10A-KB684K3-	CP10A-KF332K-	CQ10A-KF332K3-
CP10A-KB105K-	CQ10A-KB105K3-	CP10A-KF472K-	CQ10A-KF472K3-
CP10A-KC123K-	CQ10A-KC123K3-	CP10A-KF562K-	CQ10A-KF562K3-
CP10A-KC153K-	CQ10A-KC153K3-	CP10A-KF682K-	CQ10A-KF682K3-
CP10A-KC223K-	CQ10A-KC223K3-	CP10A-KF103K-	CQ10A-KF103K3-
CP10A-KC473K-	CQ10A-KC473K3-	CP10A-KF153K-	CQ10A-KF123K3-
CP10A-KC563K-	CQ10A-KC563K3-	CP10A-KF223K-	CQ10A-KF223K3-
CP10A-KC683K-	CQ10A-KC683K3-	CP10A-KF333K-	CQ10A-KF333K3-
CP10A-KC823K-	CQ10A-KC823K3-	CP10A-KF473K-	CQ10A-KF473K3-
CP10A-KC104K-	CQ10A-KC104K3-	CP10A-KF683K-	CQ10A-KF683K3-
CP10A-KC124K-	CQ10A-KC124K3-	CP10A-KF104K-	CQ10A-KF104K3-
CP10A-KC154K-	CQ10A-KC154K3-	CP10A-KF124K-	CQ10A-KF124K3-
CP10A-KC184K-	CQ10A-KC184K3-	CP10A-KF154K-	CQ10A-KF154K3-
CP10A-KC224K-	CQ10A-KC224K3-	CP10A-KF184K-	CQ10A-KF184K3-
CP10A-KC274K-	CQ10A-KC274K3-	CP10A-KF224K-	CQ10A-KF224K3-
CP10A-KC334K-	CQ10A-KC334K3-	CP10A-KF274K-	CQ10A-KF274K3-
CP10A-KC474K-	CQ10A-KC474K3-	CP10A-KF334K-	CQ10A-KF334K3-
CP10A-KC564K-	CQ10A-KC564K3-	CP10A-KF394K-	CQ10A-KF394K3-
CP10A-KC684K-	CQ10A-KC684K3-	CP10A-KF474K-	CQ10A-KF474K3-
CP10A-KC824K-	CQ10A-KC824K3-	CP10A-KF684K-	CQ10A-KF684K3-
CP10A-KC105K-	CQ10A-KC105K3-	CP10A-KG153K-	CQ10A-KG153K3-
CP10A-KE822K-	CQ10A-KE822K3-	CP10A-KG223K-	CQ10A-KG223K3-
CP10A-KE103K-	CQ10A-KE103K3-	CP10A-KG273K-	CQ10A-KG273K3-
CP10A-KE333K-	CQ10A-KE333K3-	CP10A-KG333K-	CQ10A-KG333K3-
CP10A-KE393K-	CQ10A-KE393K3-	CP10A-KG473K-	CQ10A-KG473K3-
CP10A-KE473K-	CQ10A-KE473K3-	CP10A-KG563K-	CQ10A-KG563K3-
CP10A-KE563K-	CQ10A-KE563K3-	CP10A-KG823K-	CQ10A-KG823K3-
CP10A-KE683K-	CQ10A-KE683K3-	CP10A-KG104K-	CQ10A-KG104K3-
CP10A-KE104K-	CQ10A-KE104K3-	CP10A-KG154K-	CQ10A-KG154K3-
CP10A-KE154K-	CQ10A-KE154K3-	CP10A-KG224K-	CQ10A-KG224K3-
CP10A-KE184K-	CQ10A-KE184K3-	CP10A-KG274K-	CQ10A-KG274K3-
CP10A-KE224K-	CQ10A-KE224K3-	CP10A-KG334K-	CQ10A-KG334K3-
CP10A-KE274K-	CQ10A-KE274K3-	CP10A-KG474K-	CQ10A-KG474K3-

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - AS

Preparing activity:
DLA - CC

(Project 5910-2118-06)

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